Odd toroidal grids: verification, the kernel method, and routes to $m \geq 5$

Working notes for Sid and Amrutanshu

October 26, 2026

Throughout $G = C_m \,\square\, C_n$ with m, n odd. Open problem: is G 3-choosable? (Cai–Wang–Zhu conjecture; Li–Shao–Petrov–Gordeev 2023 show $AT(G) = 4$, so the Nullstellensatz alone cannot prove it.) Amrutanshu's notes claim the case $m = 3$.

1. Novelty check

No paper since 2023 settling 3-choosability of any odd×odd toroidal grid turned up. The cited "Li–Shao–Ren, some special toroidal grids" could not be found online; since LSPG say their paper *generalizes* it and still call the odd×odd list chromatic number undetermined, it very likely does not settle $C_3 \square C_n$. Still: email the authors or check MathSciNet before submitting.

2. Independent check of the edge-deletion theorem

I computed $c_w = [x_w \prod_{v \neq w} x_v^2]\, P_{G-e}$ directly from the Lasoń / Karasev–Petrov coefficient formula (cube roots of unity at every vertex), by a column transfer over *all* 3^m column assignments, not the paper's 6-state reduction. For $m = 3$, $n \in \{3, 5, 7, 9\}$, every value of $|c_w|$ matches Theorem 1 exactly, for both column and row edges and every w, including $c_w = 0$ when w is on the cycle of e (Remark 6.2).

Beyond $m = 3$ (Remark 6.3's claim), minimum of $|c_w|$ over all w off the cycle of e:

graph	column edge	row edge
$C_5 \square C_5$	804	804
$C_5 \square C_7$	30060	30060
$C_5 \square C_9$	993855	993855
$C_5 \square C_{11}$	31835550	31835550
$C_7 \square C_5$	30060	30060
$C_7 \square C_7$ (5 positions of w sampled)	11432505	11432505

All nonzero. So for these graphs too, any 3-list assignment with an edge whose ends have disjoint lists is colourable.

Why the hand proof does not extend directly. The undamaged transfer matrix has nonzero spectrum $\{\pm i\sqrt{3}\}$ for $m = 3$ (hence $T^3 = -3T$), but six pairs $\pm i\lambda_k$ for $m = 5$ and 25 pairs for

$m = 7$. The $\pm$ pairing is what forces $c = 0$ for G itself at odd n; it persists for all m. A closed form for $m = 5$ by hand is unrealistic; a computer-assisted proof for fixed m is realistic (Section 4).

3. The kernel method (Galvin's tool)

Bondy–Boppana–Siegel: if G has a kernel-perfect orientation with $\Delta^+ \leq 2$, then G is 3-choosable. This method is independent of the Alon–Tarsi obstruction, so it is a natural candidate. Since G is 4-regular with $2mn$ edges, $\Delta^+ \leq 2$ forces $d^+(v) = d^-(v) = 2$ everywhere (Eulerian orientation).

Observation 1. *Every orientation of G with $\Delta^+ \leq 2$ contains an odd directed cycle, so Richardson's theorem never applies.*

Proof. Each weak component of an Eulerian digraph is strongly connected. If no directed cycle were odd, the parity argument of our Theorem B (distance parity from a root is well defined, every arc flips it) would make the underlying graph bipartite, but G contains the odd cycle C_m. □

So any kernel-perfect orientation must contain odd directed cycles, none of them induced. Exhaustive search over all orientations with $\Delta^+ \leq 2$ (an induced odd directed cycle has no kernel, so it rules out kernel-perfectness):

graph	orientations	no induced odd dicycle	kernel-perfect
$C_3 \square C_3$	148	12	all 12 (Galvin)
$C_3 \square C_5$	2116	0	none
$C_3 \square C_7$	33028	0	none
$C_3 \square C_9$	525316	12	all 12 (checked all 2^{27} subsets)
$C_3 \square C_{11}$	first 400000	0	none so far (partial)
$C_5 \square C_5$	143224	0	none

The 12 surviving orientations of $C_3 \square C_9$ are exactly the lifts of the 12 kernel-perfect orientations of $C_3 \square C_3$ under the covering $(r, c) \mapsto (r, c \bmod 3)$. In those, every row and column of $C_3 \square C_3$ is a transitive triangle. Lifts to $C_3 \square C_{15}$ and $C_3 \square C_{21}$ also have no induced odd directed cycle (full kernel-perfectness not checked). Lifting in both coordinates fails: all 12 lifts to $C_9 \square C_9$ contain an induced odd directed cycle (lengths 15 to 33).

Conjecture 1. *For odd $3 \leq m \leq n$, $C_m \square C_n$ has a kernel-perfect orientation with $\Delta^+ \leq 2$ if and only if $m = 3$ and $3 \mid n$.*

Remark. Consequence: the kernel method cannot reach the open case $m \geq 5$ (it already fails on $C_5 \square C_5$). Together with $AT(G) = 4$, this says both standard tools fail, which is a good motivating section for a paper built on Amrutanshu's structural method.

4. Evaluation: what to build on for $m \geq 5$

1. **Alon–Tarsi on G:** dead ($AT = 4$).
2. **Kernel method:** dead at $C_5 \square C_5$ (exhaustive).

3. **Amrutanshu's structural route** (edge-deletion AT + strip lemma + prism lemma + propagation Lemma 3.5 of [P]): the only live route. Status of each piece for $m = 5$:
   - Edge deletion: holds numerically (table in §2). Proof idea: for fixed m, c_w is a trace of transfer-matrix products; for large n the top pair $\pm i\lambda_1$ dominates, so $c_w \neq 0$ for $n \geq n_0(m)$ if the leading amplitude is nonzero, and $n < n_0$ is a finite computer check. That is a complete computer-assisted proof for $m = 5$.
   - Strip lemma: $AT(C_{\text{odd}} \square P_k) = 3$ (Kaul–Mudrock) suggests the end-multiset coefficients stay nonzero; testable now with the same code once the definitions in [P] are available.
   - Prism lemma: the triangle Hall argument has an odd-cycle analogue (an odd cycle with 2-lists is colourable unless all lists are the same 2-set), but $C_m \square K_2$ for $m \geq 5$ needs a new case analysis.
   - Lemma 3.5 of [P]: unknown to me; this is the piece to read first.

5. Suggested division of work

1. Sid: prove the kernel-method conjecture (both directions), using the parity observation above and the lift structure. Self-contained, uses the 2-kernel toolkit, becomes a section of the joint paper.

2. Sid: computer-assisted proof of the edge-deletion theorem for $m = 5$ via the dominant-eigenvalue argument.

3. Both: read [P] Lemma 3.5 and the strip lemma, then test each for $m = 5$ numerically before attempting proofs.